Assignments:
Q1. Implement following function to delete element from specified position in given array.
The function should return the updated logical size of array.
int deleteFromPosition(int[] arr, int n, int pos);

```java
//Q1. Implement following function to delete element from specified position in given array.
//The function should return the updated logical size of array.
//int deleteFromPosition(int[] arr, int n, int pos);

import java.util.Arrays;
import java.util.Scanner;

public class ResizableArray {
    public static int deleteFromPosition(int[] arr, int n, int pos) throws Exception {  1 usage
        if(pos > n) throw new Exception("Position is Greater than Logical Size");
        else{
            for(int i = pos; i < n-1; i++){
                arr[i] = arr[i+1];
            }
        }
        return n-1;
    }

    public static void main(String[] args) throws Exception {
        Scanner sc = new Scanner(System.in);
        int[] arr = {2,3,4,5,6};
        System.out.println("Array Before Deletion: ");
        System.out.println(Arrays.toString(arr));
        System.out.println("Enter position to delete from: ");
        int pos = sc.nextInt();
        int result = deleteFromPosition(arr, arr.length, pos);
        System.out.println("Array After Deletion: ");
        for(int i = 0 ; i < result; i++){
            System.out.print(arr[i]+" ");
        }
        System.out.println("\nLogical size updated to : " + result);
        sc.close();
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "
Array Before Deletion:
[2, 3, 4, 5, 6]
Enter position to delete from:
3
Array After Deletion:
2 3 4 6
Logical size updated to : 4

Process finished with exit code 0
```

Q2. Implement a Stack class using resizable array that implements the Stack interface defined in class.
Throw appropriate exception when required.

```
package org.dsa.stack;

public interface Stack {  2 usages  1 implementation
    public int pop() throws Exception;  1 usage  1 implementation
    public void push(int element) throws Exception;  1 usage  1 implementation
    public int peek() throws Exception;  1 usage  1 implementation
    public boolean isEmpty();  1 implementation
    public boolean isFull();  1 usage  1 implementation
}
```

```
package org.dsa.stack;

public class MyStack implements Stack{

    int[] arr;
    int top;
    int size = 5;
    public MyStack() {
        arr = new int[size];
        top = -1;
    }

    @Override
    public int pop() throws Exception {
        if(isEmpty()){
            throw new Exception("Stack UnderFlow/Empty");
        }
        int res = arr[top];
        top--;
        return res;
    }

    @Override
    public void push(int element) throws Exception {
        top++;
        if(isFull()) {
            size = size*2;
            int[] resized = new int[size];
            for(int i = 0 ; i < top; i++){
```

```
            resized[i] = arr[i];
        }
        arr = resized;
    }
    arr[top] = element;
}

@Override
public int peek() throws Exception {
    if(isEmpty()){
        throw new Exception("Stack UnderFlow/Empty");
    }
    return arr[top];
}

@Override
public boolean isEmpty() {
    return top==-1;
}

@Override
public boolean isFull() {
    return top==size;
}
}
```

```
package org.dsa.stack;

public class Main {
    public static void main(String[] args){
        Stack myStack = new MyStack();
        try {
            for(int i = 10; i <= 20; i++){
                myStack.push(i);
            }
            for(int i = 10; i <= 20; i++){
                System.out.println(myStack.pop());
            }
            System.out.println(myStack.peek());
```

```
    } catch (Exception e) {
        throw new RuntimeException(e);
    }




  }
}
```

Output:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.4\lib\idea_
20
19
18
17
16
15
14
13
12
11
10
Exception in thread "main" java.lang.RuntimeException Create breakpoint : java.lang.Exception: Stack UnderFlow/Empty
    at org.dsa.stack.Main.main(Main.java:16)
Caused by: java.lang.Exception Create breakpoint : Stack UnderFlow/Empty
    at org.dsa.stack.MyStack.peek(MyStack.java:40)
    at org.dsa.stack.Main.main(Main.java:13)

Process finished with exit code 1
```

Q3. Define Queue as ADT (Refer to class notes).

```java
package org.dsa.queueADT;

public interface Queue {
    public void enqueue(int element) throws Exception;
    public int dequeue() throws Exception;
    public boolean isEmpty();
    public boolean isFull();
}
```

```java
package org.dsa.queueADT;
import org.dsa.queueADT.Queue;

public class MyQueue implements Queue{

    int[] arr;
    int front;
    int rear;
    public MyQueue() {
        arr = new int[5];
        front = -1;
        rear = -1;
    }

    @Override
    public void enqueue(int element) throws Exception {
        rear++;
        if(isFull()){
            throw new Exception("Queue Full");
        }
        arr[rear] = element;
    }

    @Override
    public int dequeue() throws Exception {
        front++;
        if(isEmpty()){
            throw new Exception("Queue Empty");
        }
        return arr[front];
    }

    @Override
    public boolean isEmpty() {
        return front == rear+1;
    }
```

```java
    @Override
    public boolean isFull() {
        return rear == arr.length;
    }
}
```

```java
package org.dsa.queueADT;

import org.dsa.queueADT.Queue;

public class Main {
    public static void main(String[] args) {
        Queue myQueue = new MyQueue();
        try {
            for(int i = 10; i <= 14; i++){
                myQueue.enqueue(i);
            }
            for(int i = 10; i <= 14; i++){
                System.out.println(myQueue.dequeue());
            }
        } catch (Exception e) {
            throw new RuntimeException(e);
        }
    }
}
```

Output:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-jav
10
11
12
13
14

Process finished with exit code 0
```

Q4. Implement a Generic Queue class using array that implements the Queue interface defined in Q1.
Throw appropriate exception when required.
(Refer to class notes)

```java
package org.dsa.genericQueue;

public interface QueueG<T>{
    public void enqueue(T element) throws Exception;
    public T dequeue() throws Exception;
    public boolean isEmpty();
    public boolean isFull();
}
```

```java
package org.dsa.genericQueue;

import org.dsa.genericQueue.QueueG;

public class GenericQueue<T> implements QueueG<T> {

    T[] arr;
    int front;
    int rear;
    public GenericQueue() {
        arr = (T[]) new Object[5];
        front = -1;
        rear = -1;
    }

    @Override
    public void enqueue(Object element) throws Exception {
        rear++;
        if(isFull()){
            throw new Exception("Queue Full");
        }
        arr[rear] = (T) element;

    }

    @Override
    public T dequeue() throws Exception {
        front++;
        if(isEmpty()){
            throw new Exception("Queue Empty");
```

```
        }
        return arr[front];
    }

    @Override
    public boolean isEmpty() {
        return front == rear+1;
    }

    @Override
    public boolean isFull() {
        return rear == arr.length;
    }
}
```

```
package org.dsa.genericQueue;
public class Main {
    public static void main(String[] args) {
        QueueG<String> myQueue = new GenericQueue<>();
        try {
            for(int i = 10; i <= 14; i++){
                myQueue.enqueue("shub");
            }
            for(int i = 10; i <= 14; i++){
                System.out.println(myQueue.dequeue());
            }
        } catch (Exception e) {
            throw new RuntimeException(e);
        }
    }

}
```

OUTPUT:

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrain
shub
shub
shub
shub
shub

Process finished with exit code 0
```